

Qatar-1b

R-0265

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_171114 · created 2026-07-17T09:50:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458071.6145 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.09 +/- 0.046
Transit depth (Rp/Rs)^2	0.8 +/- 0.83 [%]
Orbital Inclination (inc)	79.46 +/- 3.12
Airmass coefficient 1 (a1)	17.3 +/- 7.34
Airmass coefficient 2 (a2)	-1.8 +/- 0.17
Scatter in the residuals of the lightcurve fit is	60.03 %
Best Comparison Star	#2 - [426, 288]
Optimal Aperture	1.48
Optimal Annulus	9.81
Transit Duration (day)	0.02 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.09 ± 0.046 vs 0.1463 (deviation 0.27σ)
Duration fit vs published	0.02 d vs 0.0692 d (deviation 0.44σ)
Mid-time O–C	-0.4 min
Depth SNR	0.4
Residual scatter	60.03 %

Light curve

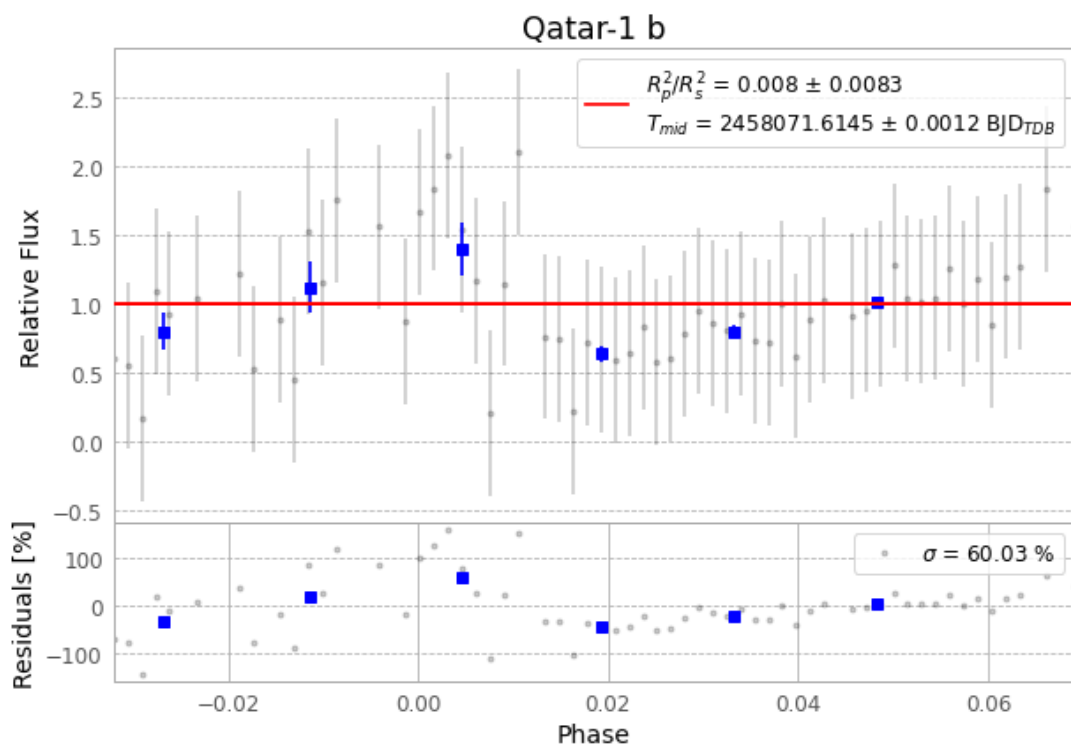


Figure 1 — FinalLightCurve_Qatar-1 b_2017-11-13.png (SHA-256 2a3c2b794f1ecab3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [378, 345], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 70db7d71ebf4b8de...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 7dbc7ef

Data provenance

70 FITS frames (46 MB), Qatar-1171114013612.FITS → Qatar-1171114050312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-171114.log (2d2bf5decb36...), FinalLightCurve_Qatar-1 b_2017-11-13.png (2a3c2b794f1e...), FinalLightCurve_Qatar-1 b_2017-11-13.csv (bf0740322f8f...)

Compute resources

Wall time 230.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 09:50Z.